1. Install nginx and run nginx on port number 81.

Step 1: yum install nginx -y (nginx will install on AWS server)

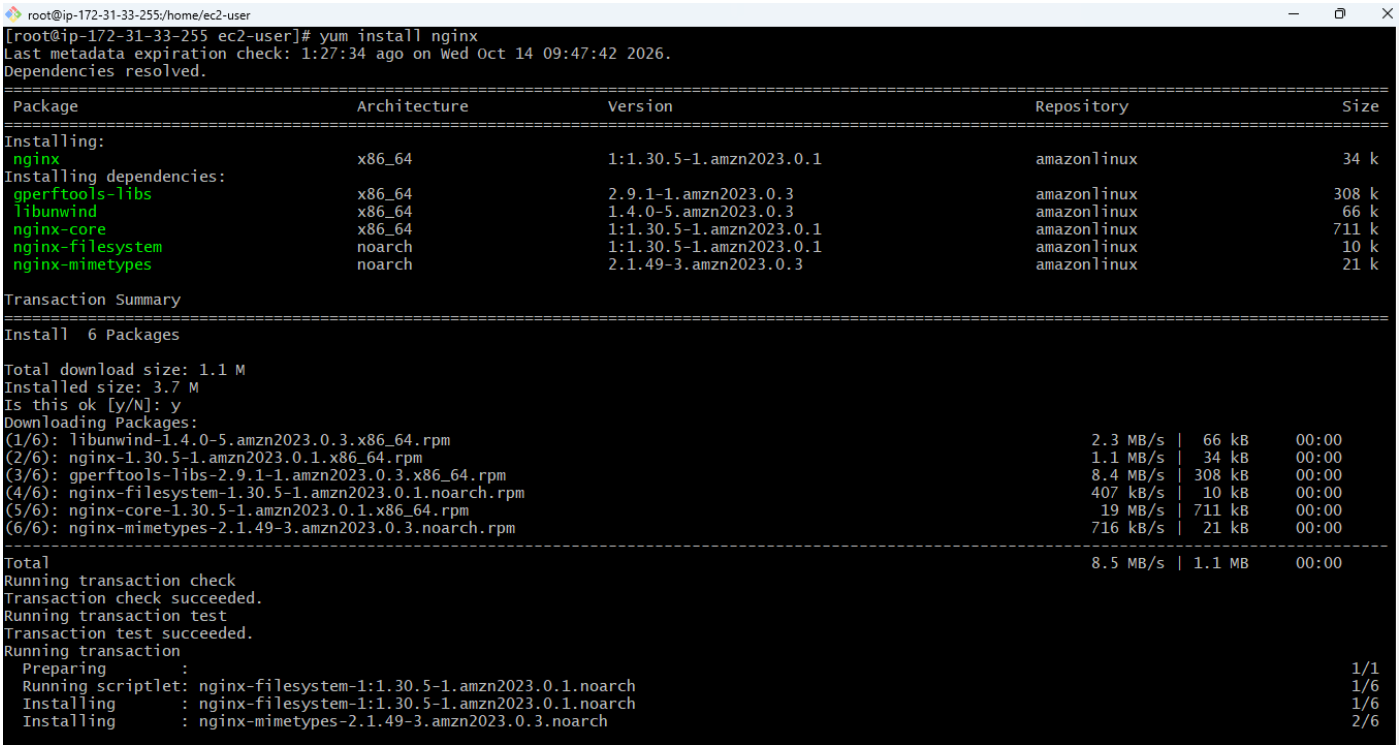
Step 2: systemctl start nginx (nginx service will start)

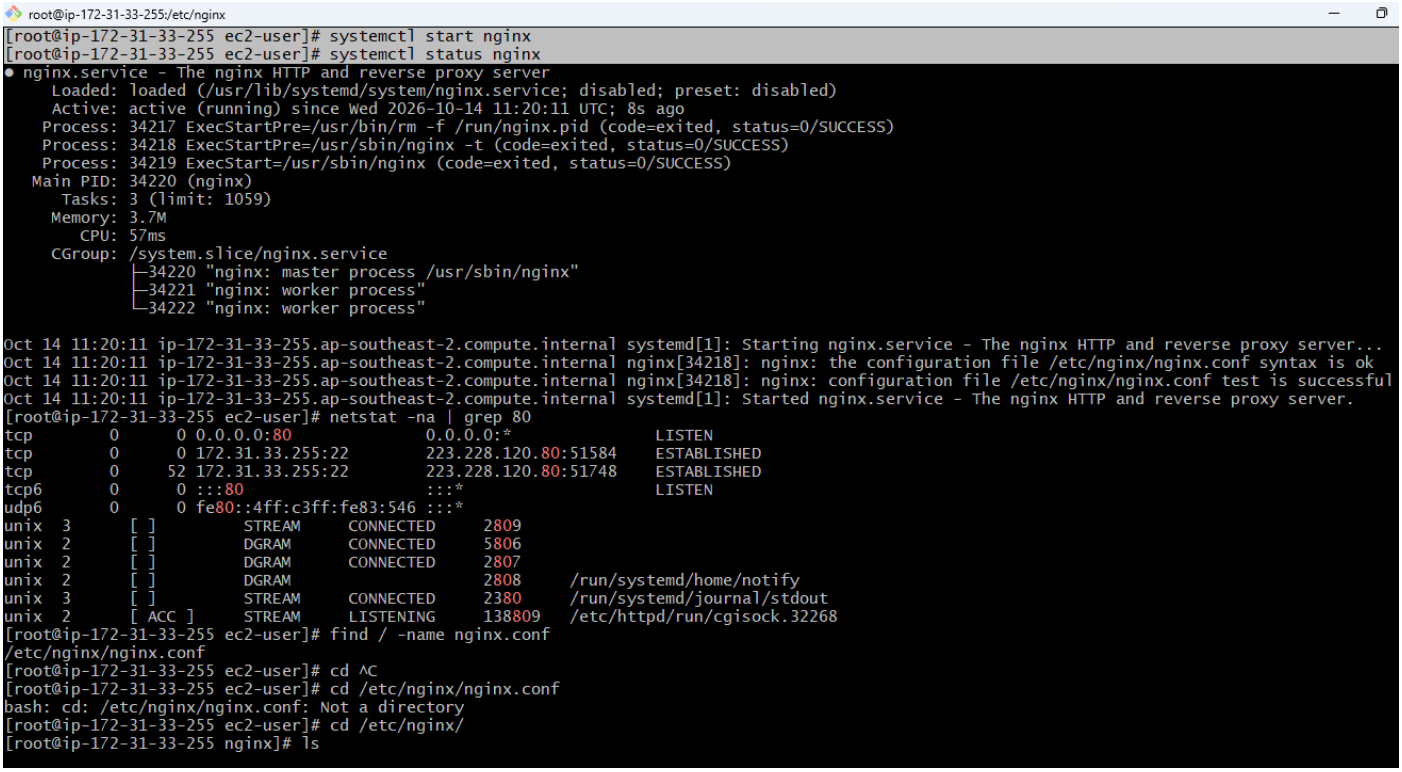
Step 3: systemctl status nginx (Check the status Active or inactive)

Step 4: netstat -na | grep 80 (Check the port 80 listening)

Step 5: go to chrome or firefox type (public ipv4:80) and webpage shows “Welcome to Nginx”

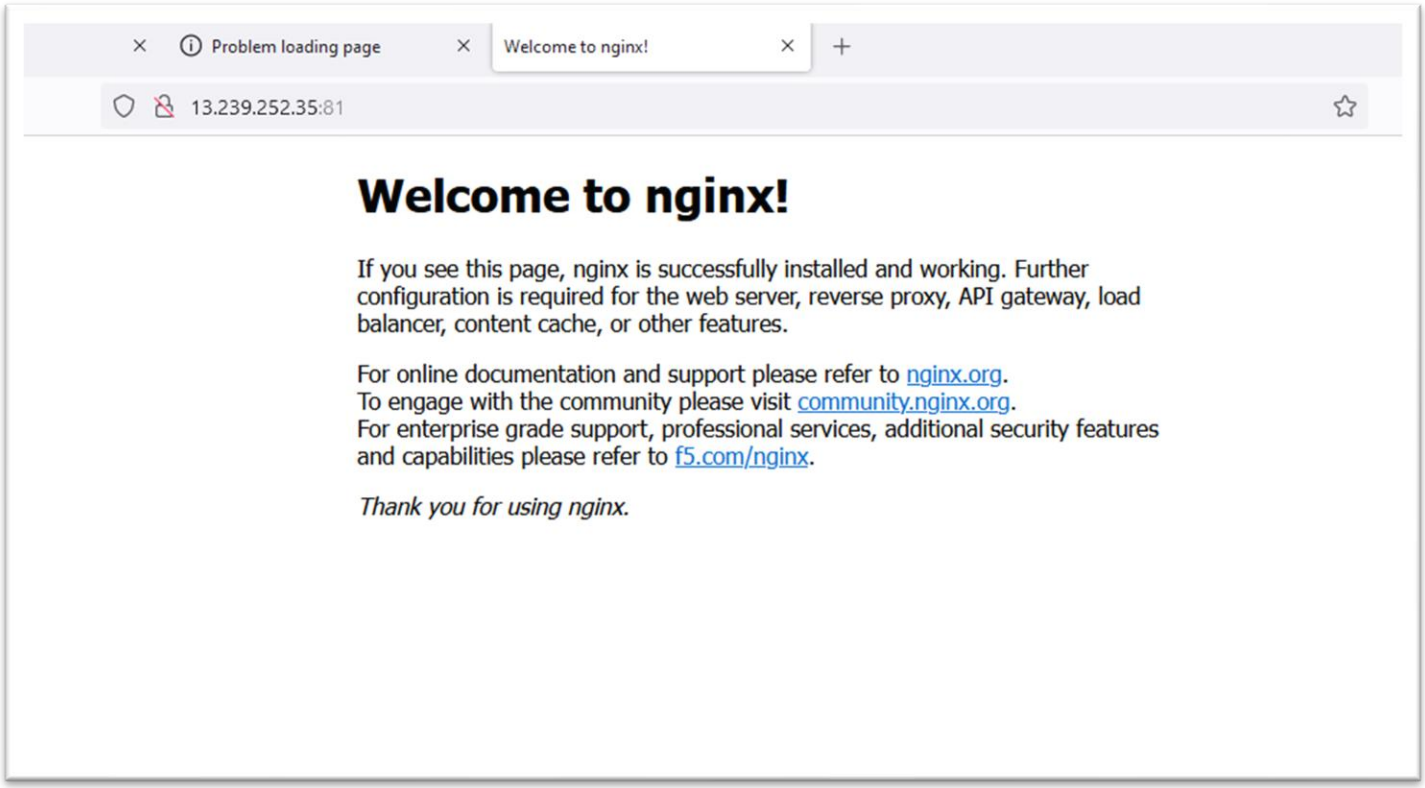
Step 6: find / -name nginx.conf (where you have to change port to 81)





```
[root@ip-172-31-33-255 ec2-user]# netstat -na | grep 80
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN
tcp        0      0 172.31.33.255:22        223.228.120.80:51584    ESTABLISHED
tcp        0     52 172.31.33.255:22        223.228.120.80:51748    ESTABLISHED
tcp6       0      0 :::80                   :::*                    LISTEN
udp6       0      0 fe80::4ff:c3ff:fe83:546 :::*
unix  3      [ ]         STREAM     CONNECTED     2809
unix  2      [ ]         DGRAM      CONNECTED     5806
unix  2      [ ]         DGRAM      CONNECTED     2807
unix  2      [ ]         DGRAM                    2808     /run/systemd/home/notify
unix  3      [ ]         STREAM     CONNECTED     2380     /run/systemd/journal/stdout
unix  2      [ ACC ]     STREAM     LISTENING     138809   /etc/httpd/run/cgisock.32268
[root@ip-172-31-33-255 ec2-user]# find / -name nginx.conf
/etc/nginx/nginx.conf
[root@ip-172-31-33-255 ec2-user]# cd ^C
[root@ip-172-31-33-255 ec2-user]# cd /etc/nginx/nginx.conf
bash: cd: /etc/nginx/nginx.conf: Not a directory
[root@ip-172-31-33-255 ec2-user]# cd /etc/nginx/
[root@ip-172-31-33-255 nginx]# ls
conf.d        fastcgi.conf.default    koi-utf     mime.types.default  scgi_params          uwsgi_params.default
default.d     fastcgi_params          koi-win     nginx.conf          scgi_params.default  win-utf
fastcgi.conf  fastcgi_params.default  mime.types  nginx.conf.default  uwsgi_params
[root@ip-172-31-33-255 nginx]# cat nginx.conf
# For more information on configuration, see:
#   * Official English Documentation: http://nginx.org/en/docs/
#   * Official Russian Documentation: http://nginx.org/ru/docs/

user nginx;
worker_processes auto;
```

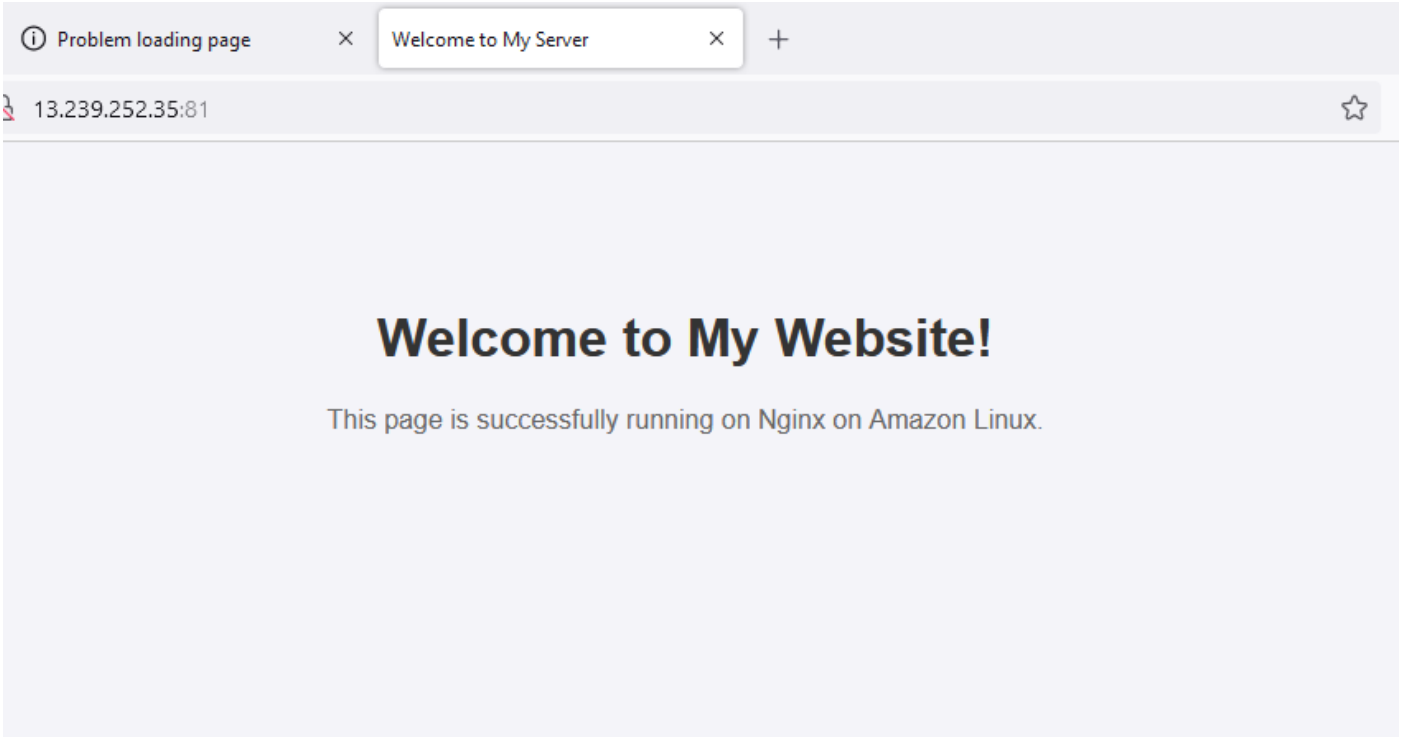


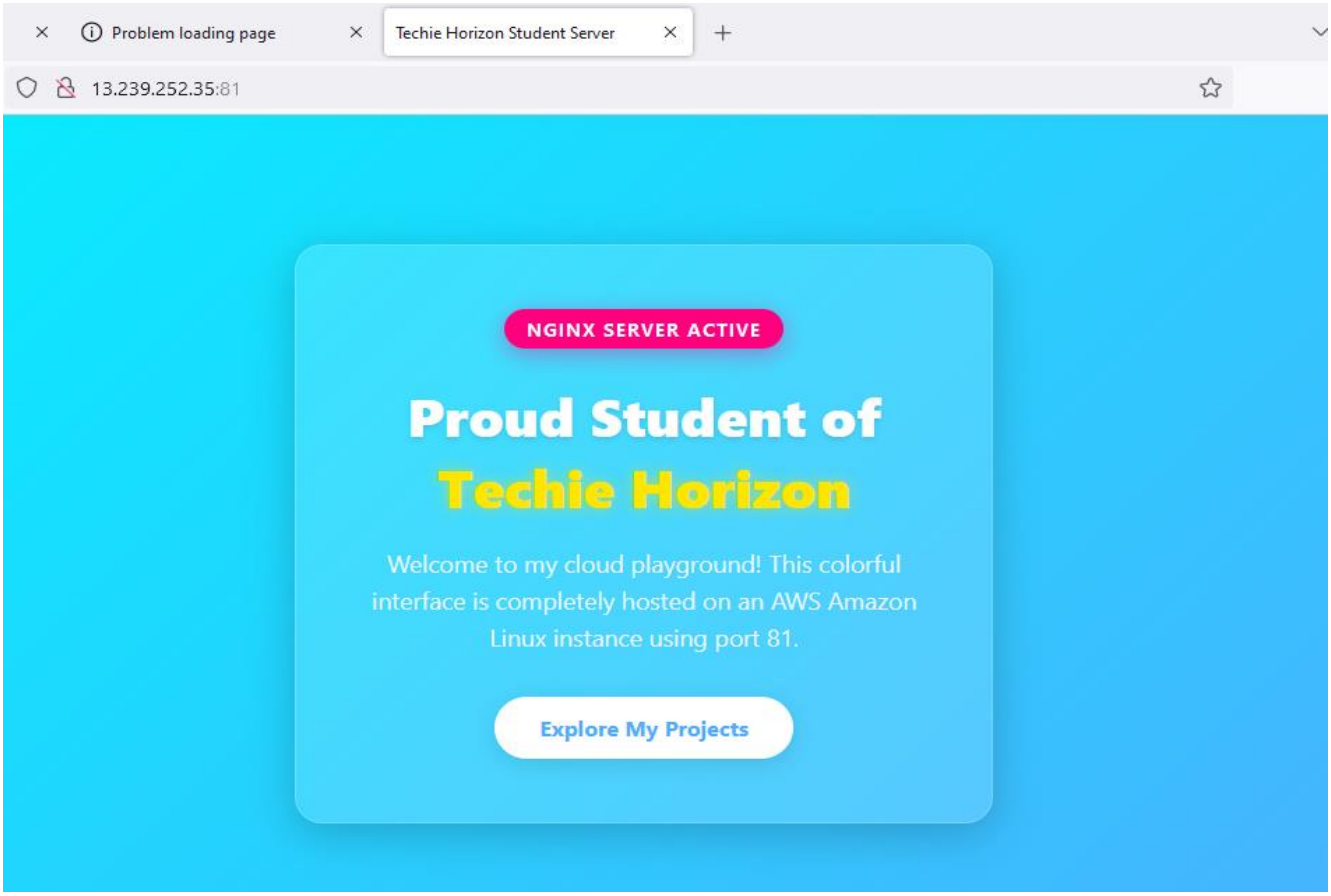
2. Deploy a sample index.html file on nginx.

Step 1: Cd /usr/share/nginx/html (There will be an existing index.html)

Step 2: vi index.html (edited the content in file)

Step 3: goto chrome or firefox enter (publicIP4 : 81) you will get webpage with content





3. Install Apache and run Apache on port number 82

Step 1: yum install httpd -y (Apache will install on AWS server)

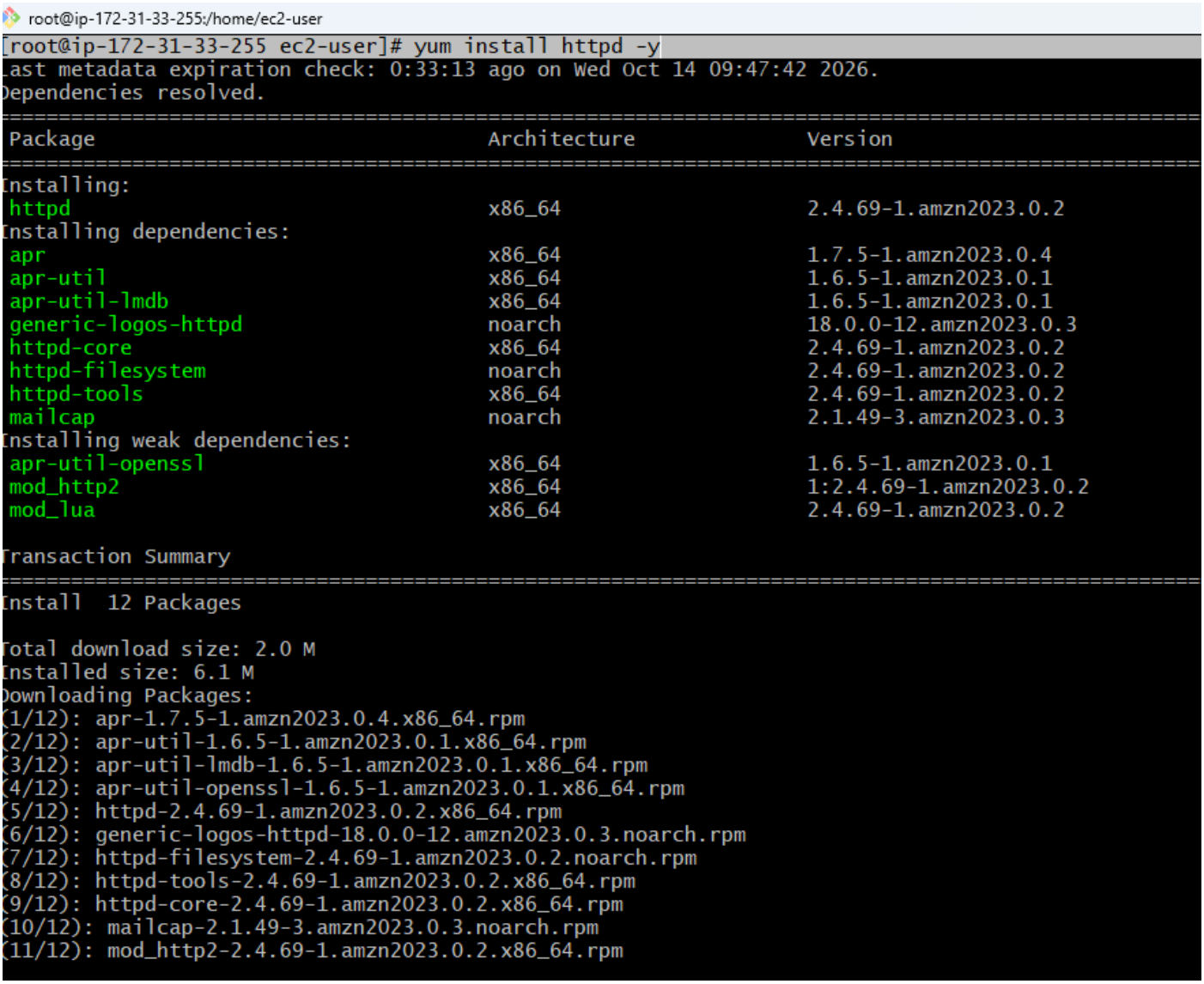
Step 2: systemctl start httpd (Apache service will start)

Step 3: systemctl status httpd (Check the status Active or inactive)

Step 4: netstat -na | grep 80 (Check the port 80 listening)

Step 5: go to chrome or firefox type (public ipv4:80) and webpage shows “it works”

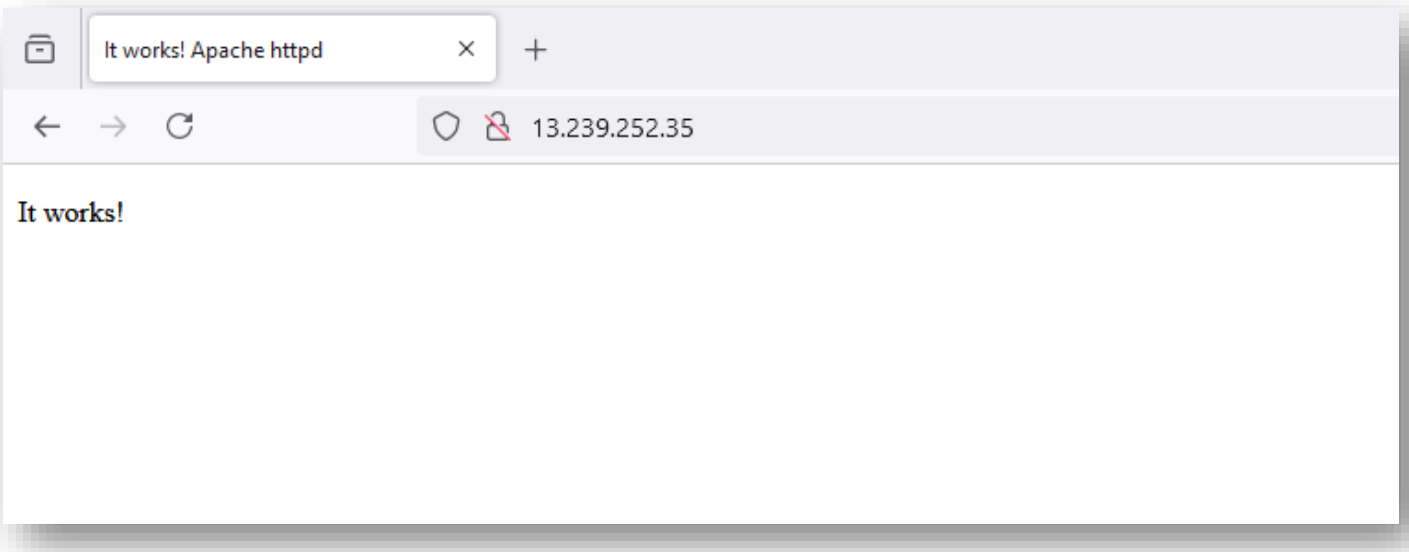
Step 6: find / -name httpd.conf (where you have to change port to 82)



```
[root@ip-172-31-33-255 ec2-user]# systemctl start httpd
[root@ip-172-31-33-255 ec2-user]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabl
     Active: active (running) since Wed 2026-10-14 10:21:48 UTC; 12s ago
       Docs: man:httpd.service(8)
   Main PID: 29634 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes serv
      Tasks: 177 (limit: 1059)
     Memory: 14.0M
        CPU: 90ms
     CGroup: /system.slice/httpd.service
             ├─29634 /usr/sbin/httpd -DFOREGROUND
             ├─29657 /usr/sbin/httpd -DFOREGROUND
             ├─29659 /usr/sbin/httpd -DFOREGROUND
             ├─29660 /usr/sbin/httpd -DFOREGROUND
             └─29661 /usr/sbin/httpd -DFOREGROUND

Oct 14 10:21:47 ip-172-31-33-255.ap-southeast-2.compute.internal systemd[1]: Startin
Oct 14 10:21:48 ip-172-31-33-255.ap-southeast-2.compute.internal systemd[1]: Started
Oct 14 10:21:48 ip-172-31-33-255.ap-southeast-2.compute.internal httpd[29634]: Serve
[root@ip-172-31-33-255 ec2-user]#
```

```
[root@ip-172-31-33-255 ec2-user]# netstat -na | grep 80
tcp        0      0 172.31.33.255:45158     169.254.169.254:80      TIME_WAIT
tcp        0      0 172.31.33.255:45178     169.254.169.254:80      TIME_WAIT
tcp        0      0 172.31.33.255:45180     169.254.169.254:80      TIME_WAIT
tcp6       0      0 :::80                   :::*                    LISTEN
udp6       0      0 fe80::4ff:c3ff:fe83:546 :::*
unix  3      [ ]         STREAM     CONNECTED     2809
unix  2      [ ]         DGRAM      CONNECTED     5806
unix  2      [ ]         DGRAM      CONNECTED     2807
unix  2      [ ]         DGRAM                    2808     /run/systemd/home/notify
unix  3      [ ]         STREAM     CONNECTED     2380     /run/systemd/journal/stdout
[root@ip-172-31-33-255 ec2-user]#
```

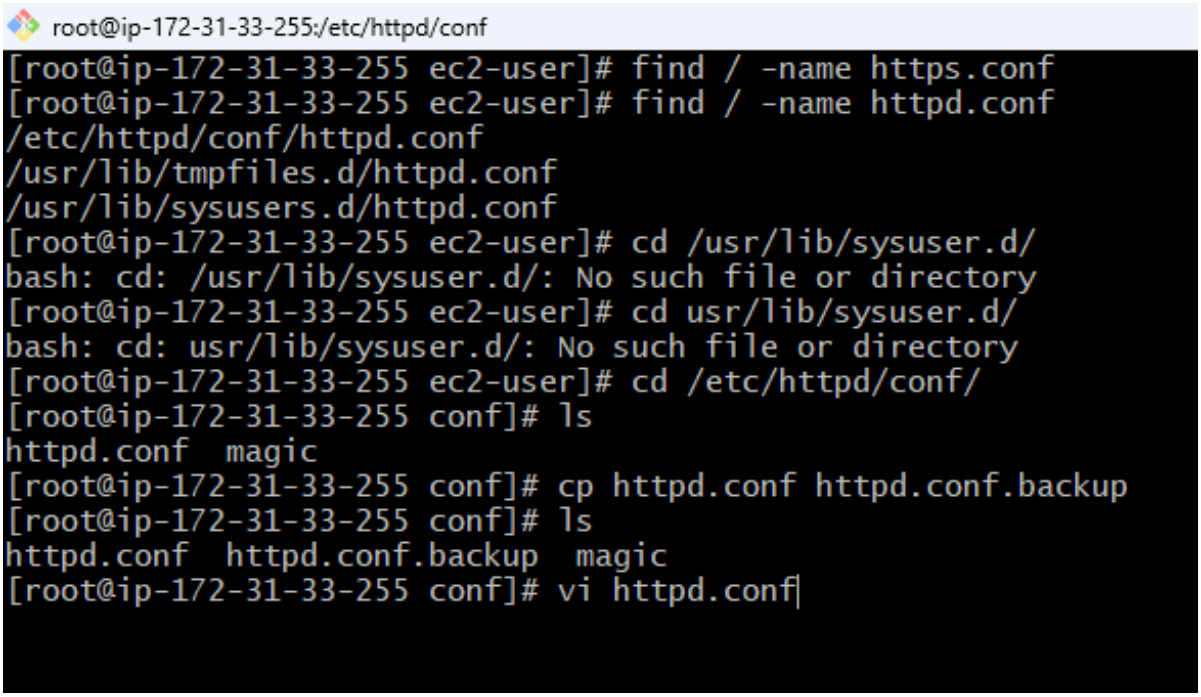


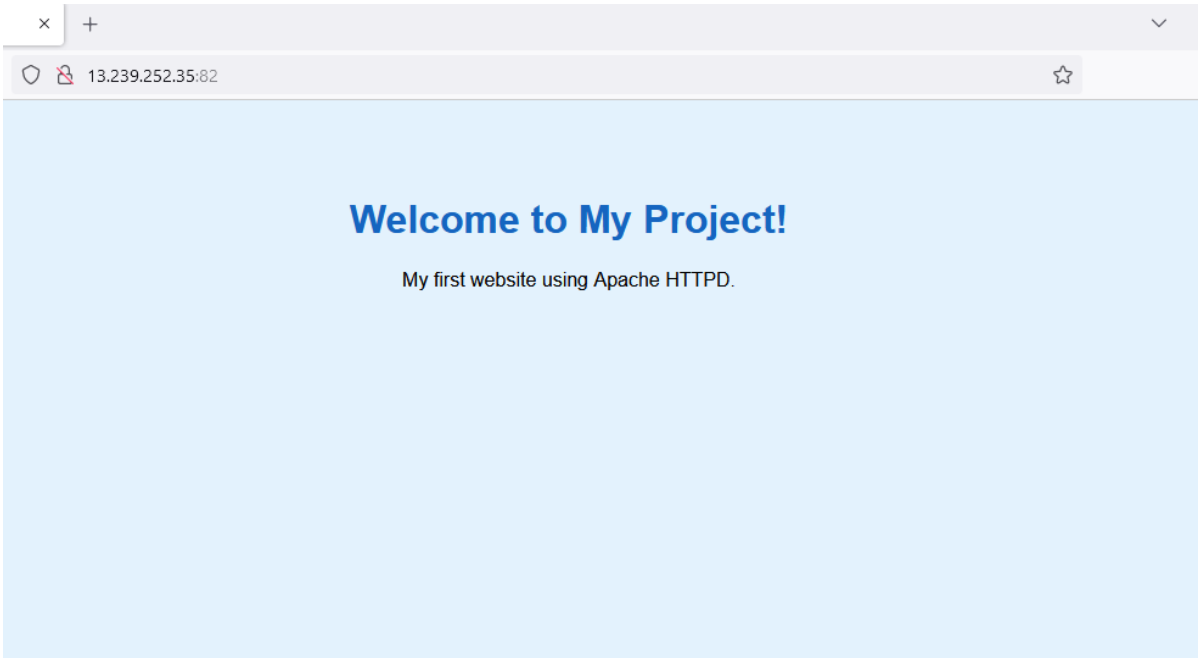
4. Deploy a sample index.html file on Apache.

Step 1: Cd /var/www/html (created a file using touch index.html)

Step 2: vi index.html (edited the content in file)

Step 3: goto chrome or firefox enter (publicIP4 : 82) you will get webpage with content





5. Install Apache tomcat on port number 8082

Step 1: yum install tomcat10 (tomcat10 will install on AWS server)

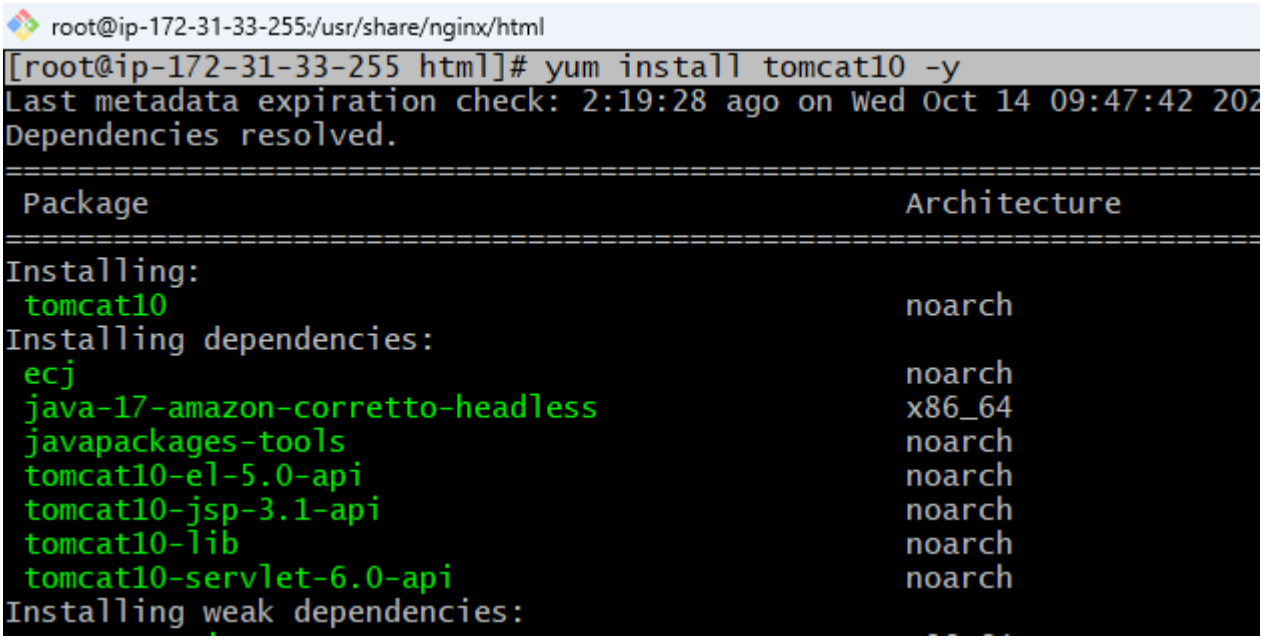
Step 2: systemctl start tomcat10 (tomcat10 service will start)

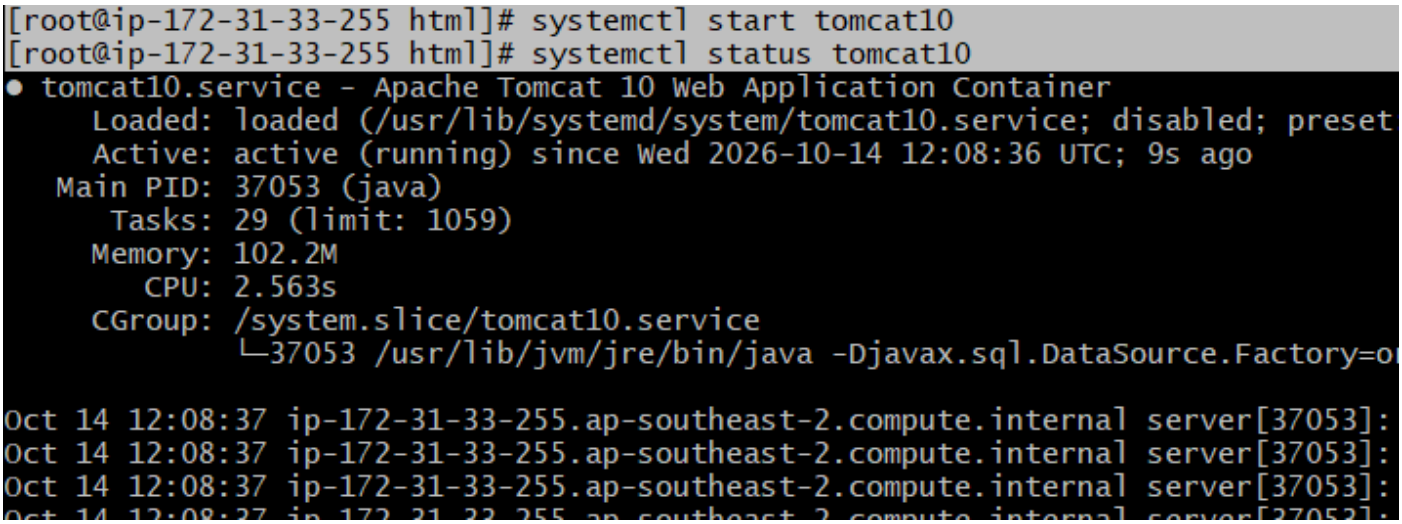
Step 3: systemctl status tomcat10 (Check the status Active or inactive)

Step 4: netstat -na | grep 8080 (Check the port 8080 listening)

Step 5: go to chrome or firefox type (public ipv4:8080) and webpage shows “tomcat webpage”

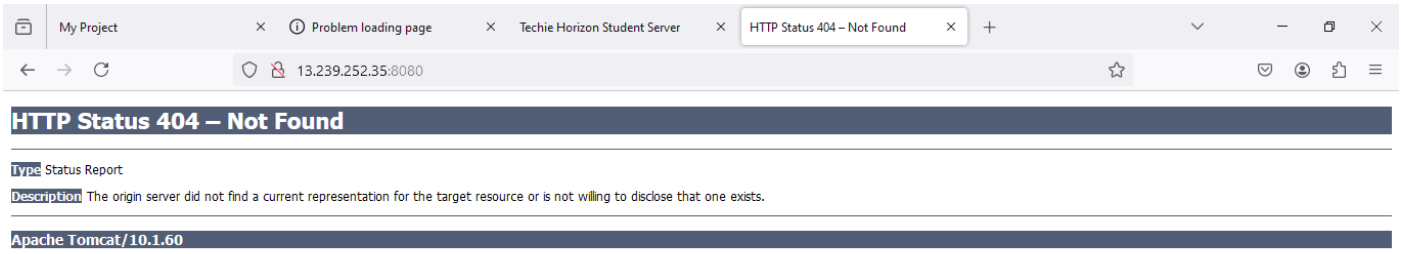
Step 6: find / -name server.xml (where you have to change port to 8080 to 8082)







Public IPv4 : 8080



Public IPv4 : 8082



6. Deploy a sample app on webapps

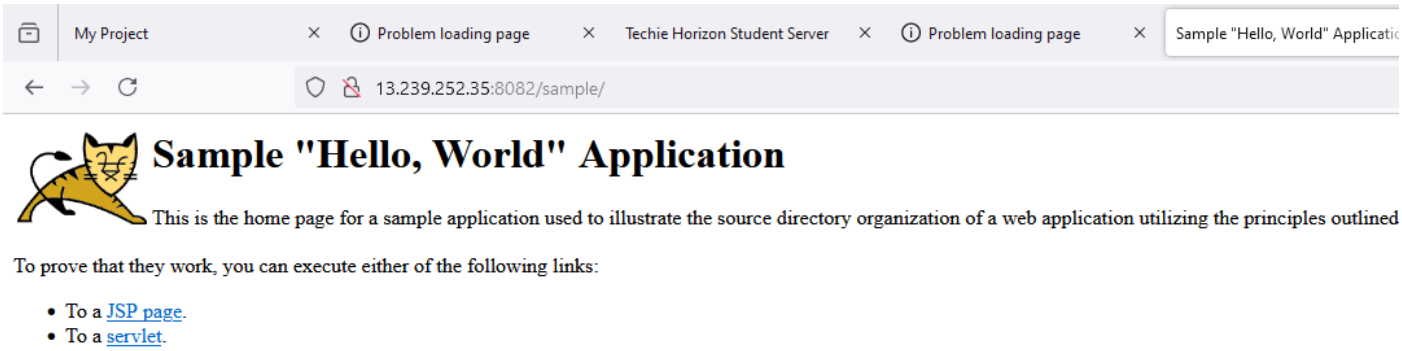
Step 1: cd /usr/share/tomcat10/webapps/ (go to webapps directory)

Step 2: go to tomcat side and copy sample.war link

Step 3: in terminal wget and paste link you will get sample.war file

Step 4: extract that file with jar -xvf sample.war

Step 5: go to chrome or firefox (publicIP4 : 8082/sample/

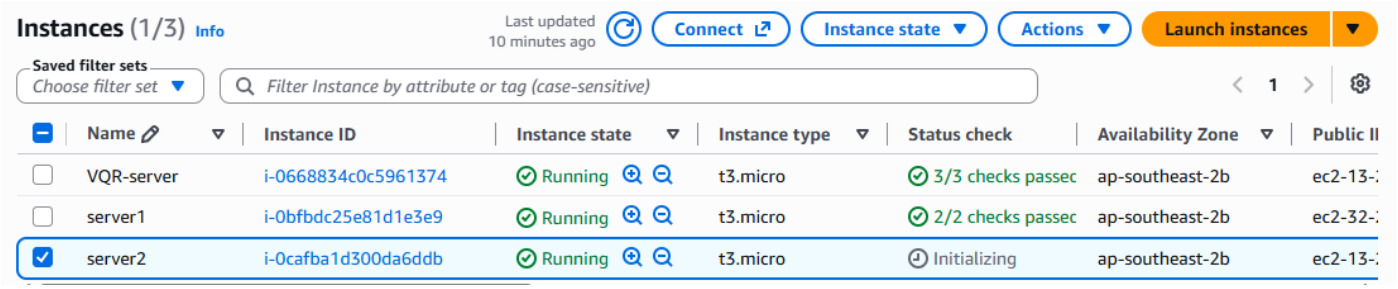


7. Create a tomcat.service file for tomcat.

```
Oct 14 14:21:26 ip-172-31-33-255.ap-southeast-2.compute.internal systemd[1]: Stopped tomcat.service - Apache Tomcat Web Application Server.
[root@ip-172-31-33-255 system]# systemctl start tomcat
Job for tomcat.service failed because the control process exited with error code.
See "systemctl status tomcat.service" and "journalctl -xeu tomcat.service" for details.
[root@ip-172-31-33-255 system]# systemctl status tomcat
● tomcat.service - Apache Tomcat Web Application Server
     Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: disabled)
     Active: activating (auto-restart) (Result: exit-code) since Wed 2026-10-14 14:23:39 UTC; 3s ago
    Process: 44582 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=203/EXEC)
        CPU: 1ms
[root@ip-172-31-33-255 system]# systemctl status tomcat
● tomcat.service - Apache Tomcat Web Application Server
     Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: disabled)
     Active: activating (auto-restart) (Result: exit-code) since Wed 2026-10-14 14:23:39 UTC; 9s ago
    Process: 44582 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=203/EXEC)
        CPU: 1ms
[root@ip-172-31-33-255 system]# systemctl enable tomcat
[root@ip-172-31-33-255 system]# systemctl status tomcat
● tomcat.service - Apache Tomcat Web Application Server
     Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: disabled)
     Active: activating (auto-restart) (Result: exit-code) since Wed 2026-10-14 14:23:59 UTC; 6s ago
        CPU: 1ms
[root@ip-172-31-33-255 system]#
```

8. Configure HA Proxy server

Created 2 instance server1 and server2 and using my existing instance as HA proxy



Server 1

installed httpd and running on port 80

```
root@ip-172-31-44-40:/
Complete!
[root@ip-172-31-44-40 /]# yum install httpd
Last metadata expiration check: 0:00:26 ago on Wed Oct 14 15:48:32 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture            Version
================================================================================
Installing:
 httpd                   x86_64                  2.4.69-1.amzn2023.0.2
Installing dependencies:
 apr                     x86_64                  1.7.5-1.amzn2023.0.4
 apr-util                x86_64                  1.6.5-1.amzn2023.0.1
 apr-util-lmdb           x86_64                  1.6.5-1.amzn2023.0.1
 generic-logos-httpd     noarch                  18.0.0-12.amzn2023.0.3
 httpd-core              x86_64                  2.4.69-1.amzn2023.0.2
 httpd-filesystem        noarch                  2.4.69-1.amzn2023.0.2
 httpd-tools             x86_64                  2.4.69-1.amzn2023.0.2
 libbrotli               x86_64                  1.1.0-1.amzn2023.0.1
 mailcap                 noarch                  2.1.49-3.amzn2023.0.3
Installing weak dependencies:
 apr-util-openssl        x86_64                  1.6.5-1.amzn2023.0.1
 mod_http2               x86_64                  1:2.4.69-1.amzn2023.0.2
 mod_lua                 x86_64                  2.4.69-1.amzn2023.0.2

Transaction Summary
================================================================================
Install  13 Packages

Total download size: 2.3 M
Installed size: 6.9 M
Is this ok [y/N]: y
Downloading Packages:
(1/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm
(2/13): apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64.rpm
(3/13): apr-util-1.6.5-1.amzn2023.0.1.x86_64.rpm
(4/13): apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64.rpm
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm
(6/13): httpd-2.4.69-1.amzn2023.0.2.x86_64.rpm
(7/13): httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch.rpm
(8/13): httpd-core-2.4.69-1.amzn2023.0.2.x86_64.rpm
(9/13): httpd-tools-2.4.69-1.amzn2023.0.2.x86_64.rpm
(10/13): libbrotli-1.1.0-1.amzn2023.0.1.x86_64.rpm
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm
(12/13): mod_lua-2.4.69-1.amzn2023.0.2.x86_64.rpm
(13/13): mod_http2-2.4.69-1.amzn2023.0.2.x86_64.rpm
--------------------------------------------------------------------------------
Total
Running transaction check
```

```
[root@ip-172-31-44-40 etc]# cat host
cat: host: No such file or directory
[root@ip-172-31-44-40 etc]# cat hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
[root@ip-172-31-44-40 etc]# vi hosts
[root@ip-172-31-44-40 etc]# cat hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
13.239.252.35 load_balancer
[root@ip-172-31-44-40 etc]#
```

root@ip-172-31-36-128:/home/ec2-user

```
Complete!
[root@ip-172-31-36-128 ec2-user]# yum install nginx -y
Last metadata expiration check: 0:44:24 ago on Wed Oct 14 15:53:06 2026.
Dependencies resolved.
================================================================================
 Package                                 Architecture                      Versi
================================================================================
Installing:
 nginx                                   x86_64                            1:1.3
Installing dependencies:
 generic-logos-httpd                     noarch                            18.0.
 gperftools-libs                         x86_64                            2.9.1
 libunwind                               x86_64                            1.4.0
 nginx-core                              x86_64                            1:1.3
 nginx-filesystem                        noarch                            1:1.3
 nginx-mimetypes                         noarch                            2.1.4

Transaction Summary
================================================================================
Install  7 Packages

Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm
(2/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm
(3/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm
(4/7): nginx-1.30.5-1.amzn2023.0.1.x86_64.rpm
(5/7): nginx-core-1.30.5-1.amzn2023.0.1.x86_64.rpm
(6/7): nginx-filesystem-1.30.5-1.amzn2023.0.1.noarch.rpm
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm
--------------------------------------------------------------------------------
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
```

```
[root@ip-172-31-36-128 etc]# vi hosts
[root@ip-172-31-36-128 etc]# cat hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost6 localhost6.localdomain6
13.239.252.35 load_balancer
[root@ip-172-31-36-128 etc]#
```

```
[ec2-user@ip-172-31-44-40 ~]$
[ec2-user@ip-172-31-44-40 ~]$ ping load_balancer -c 4
PING load_balancer (13.239.252.35) 56(84) bytes of data.
```

```
13.239.252.35 load_balancer
[root@ip-172-31-36-128 etc]# ping load_balancer -c 4
PING load_balancer (13.239.252.35) 56(84) bytes of data.
```

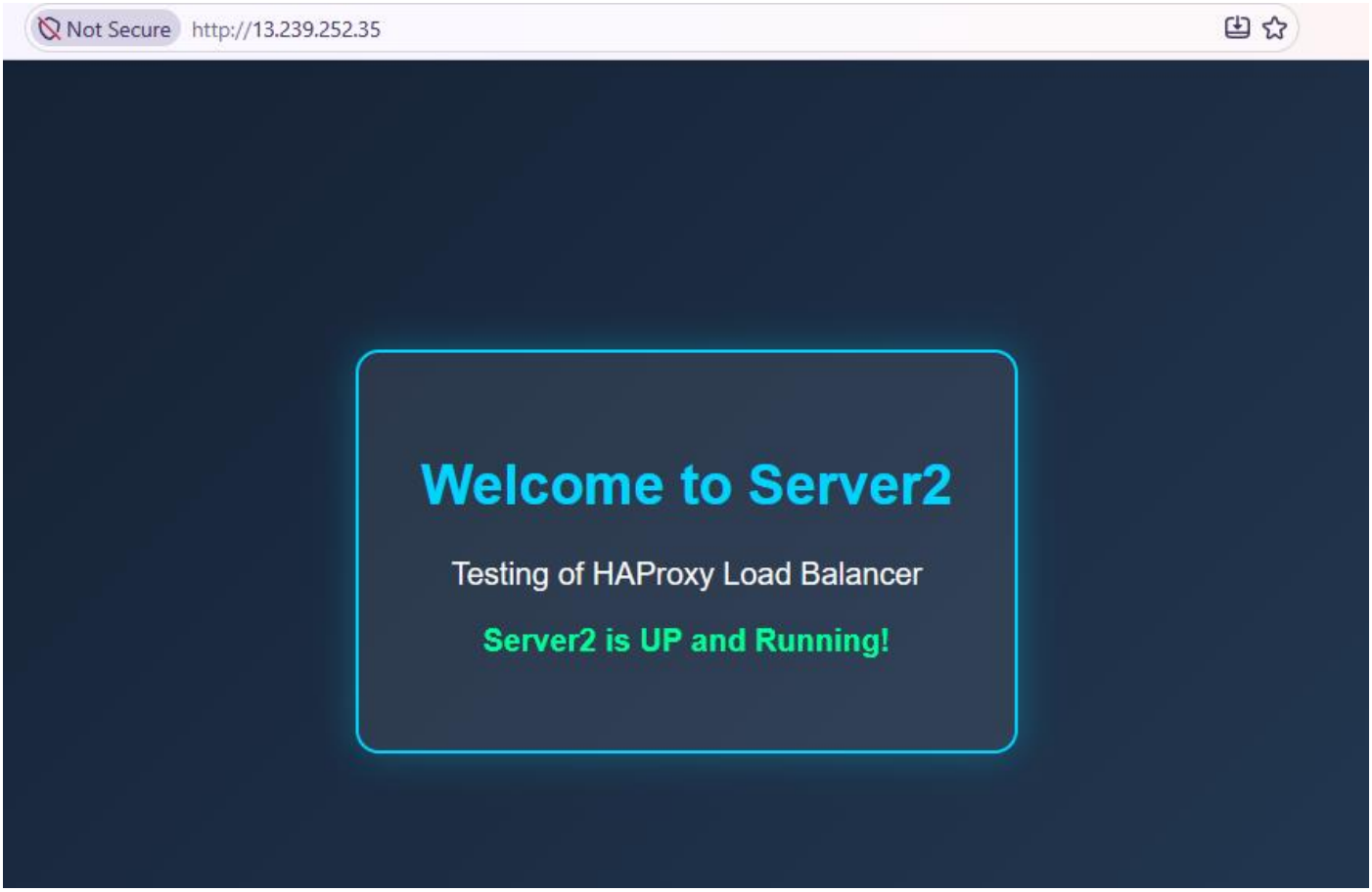
```
[root@ip-172-31-33-255 ec2-user]# yum install haproxy -y
Last metadata expiration check: 7:05:12 ago on Wed Oct 14 09:47:42 2026.
Dependencies resolved.
================================================================================
 Package                         Architecture                    Version
================================================================================
Installing:
 haproxy                         x86_64                          3.0.23-2.a

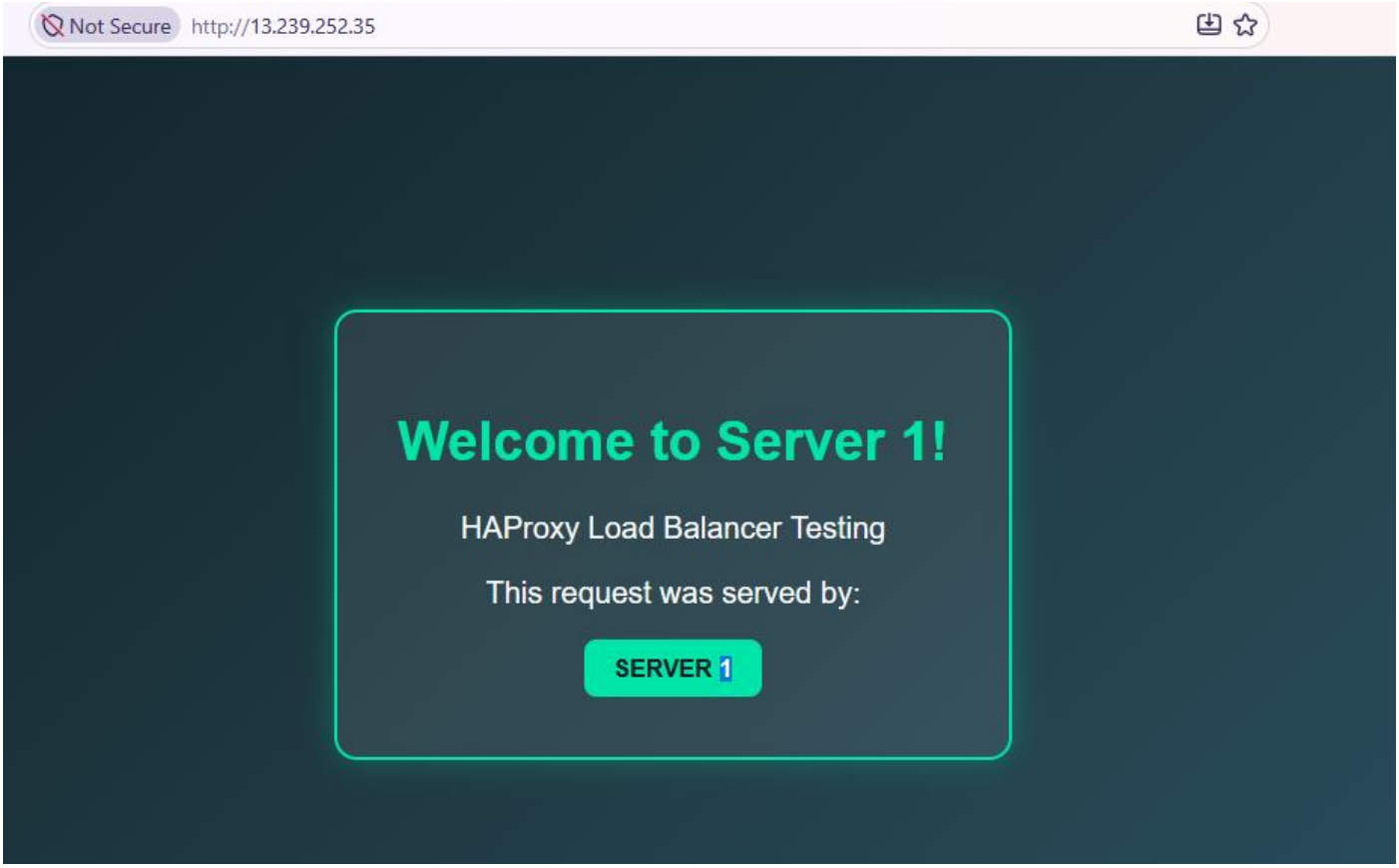
Transaction Summary
================================================================================
Install  1 Package

Total download size: 2.7 M
Installed size: 8.5 M
Downloading Packages:
haproxy-3.0.23-2.amzn2023.0.1.x86_64.rpm
--------------------------------------------------------------------------------
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :
  Running scriptlet: haproxy-3.0.23-2.amzn2023.0.1.x86_64
  Installing       : haproxy-3.0.23-2.amzn2023.0.1.x86_64
  Running scriptlet: haproxy-3.0.23-2.amzn2023.0.1.x86_64
  Verifying        : haproxy-3.0.23-2.amzn2023.0.1.x86_64

Installed:
  haproxy-3.0.23-2.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-33-255 ec2-user]#
```




Not Secure
http://13.239.252.35
Welcome to Server 1!
HAProxy Load Balancer Testing
This request was served by:
SERVER 1